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Problems on Races

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Question 1

Akshat runs 100 m in 20 s and Akshara runs the same distance in 10 sec. By what distance will Akshara beat Akshat in a 100-m race?

A) 50 m

B) 25 m

C) 40 m

D) 60 m

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Question 2

In a 100-m race, A beats B by 10 m and B beats C by 10 m. Find the distance by which A beats C.

A) 10 m

B) 20 m

C) 19 m

D) 11 m

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Question 3

In a 250 m race, the ratio of the speed of Ram and Rama is 3 : 2. If Ram has a start of 70 m, then Ram wins by

A) 130 m

B) 100 m

C) 120 m

D) 120 m

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Question 4

Sandeepa is 100% faster than Manisha. In a race, Sandeepa gave Manisha a head start of 200 m. Both of them finished the race simultaneously. Find the length of the race in metres.

A) 200

B) 400

C) 300

D) 500

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Question 5

Three friends A, B and C decide to run around a circular track. They start at the same time and run in the same direction. A is the quickest and when A finishes a lap, it is seen that C is as much behind B as B is behind A. When A completes 3 laps, C is the exact same position on the circular track as B was when A finished 1 lap. Find the ratio of the speeds of A, B and C?

A) 5:4:2

B) 4:3:2

C) 5:4:3

D) 3:2:1

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Question 6

A and B stand at distinct points of a circular race track of length 120m. They run at speeds of a m/s and b m/s respectively. They meet for the first time 16 seconds after they start the race and for the second time 40 seconds from the time they start the race. Now, if B had started in the opposite direction to the one he had originally started, they would have meet for the first time after 40 seconds. If B is quicker than A, find B's speed.

A) 5:4:2

B) 4:3:2

C) 5:4:3

D) 3:2:1

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Question 7

Akbar, Birbal and Chanakya run around a circular track of length 500 m. Akbar and Birbal run with the speeds of 15 m/s and 20 m/s in the same direction respectively and Chanakya being very intelligent run in che opposite direction with a speed of 25 m/s. If all three Of them start at the same time,

A) Akbar meets Chanakya more frequently than Birbal does

B) Akbar and Chanakya meets as frequently as Birbal and Chanakya

C) Akbar meets Birbal least frequently

D) Nothing can be concluded

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Question 8

Arun and Barun run with the speeds of 30 m/s and 20 m/s around a circular track of 600 m.They participate in a 3000 m race. What is the distance covered by Arun when he pass Barun for the 5th time?

A) 2200 m

B) 2250 m

C) 2850 m

D) None of these

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Question 9

Akkal and Bakkal are running on a circular track of radius 175 metres. Akkal can complete a round in 100 seconds and the speed of Bakkal is twice the speed of Akkal. They started simultaneously towards each other from two points 350 metres diametrically opposite on the circular path. If they first meet at a point they called it love point, which is between the two points P and Q from where they have started their race, after how much time from the start do they meet at love point for the third dine?

A) 218 x 2/5 s

B) 216 x 2/3 s

C) 221 s

D) None of these

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Question 10

In a 200 m race, Amar beats Akbar by 20 m and Anthony by 50 m. If Akbar and Anthony are running a 100 m race with the same speed, find the distance by which Akbar will beat Anthony.

A) 83.33 m

B) 16.66 m

C) 11.11 m

D) 88.89 m

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Question 11

A and B runs around a circular track. A beats B by one round or 10 minutes. In this race, they had completed 4 rounds. If the race was only of one round, find the A's time over the course :

A) 8 minutes

B) 7.5 minutes

C) 12.5 minutes

D) 12 minutes

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Question 12

A, B and C participated in a race. A covers the same distance in 49 steps, as B covers in 50 steps and C in 51 steps. A takes 10 steps in the same time as B takes 9 steps and C takes 8 steps. Who is the winner of the race?

A) A

B) B

C) C

D) can't be determined

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Question 13

Two friends A and B run around a circular track of length 510 metres, starting from the same point, simultaneously and in the same direction. A who runs faster laps B in the middle of the 5th round. If A and B were to run a 3 km race long race, how much start, in terms of distance, should A give B so that they finish the race in a dead heat?

A) 545.45 metres

B) 666.67 metres

C) 857.14 metres

D) Can't be determined

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Question 14

Twenty six men - 1,2,3,....25 and 26 participate in 10km running race on a circular track of length 100m. All of them start at the same time, from the same point and run in the same direction. Their speeds, taken in the order, are in increasing AP. The time taken by 26 to meet 1, for the first time after they start is 20 sec and the time taken by 13 to complete the race is 52 minutes and 5 seconds. Find the time taken (in seconds), for all the twenty six men to meet for the first time at the starting point.

A) 1000

B) 500

C) 625

D) 400

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Question 15

Two motorists Anil and Sunil are practicing with two different sports car; Ferrari and McLaren, on the circular racing track, for the car racing tournament to be held next month. Both Anil and Sunil start from the same point on the circular track. Anil completes one round of the track in 1 min and Sunil takes 2 min to complete a round. While Anil maintains speed for all the rounds, Sunil halves his speed after the completion of each round. How many times Anil and Sunil will meet between 6th round and 9th round of Sunil (6th and 9th round is excluded)? Assume that the speed of Sunil remains steady throughout each round and changes only after the completion of that round

A) 382

B) 347

C) 260

D) None of these

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THANK YOU

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